

the open source community, online forums and even via experts. Round-the-clock support is available for MariaDB via enterprise subscription especially for mission-critical production systems.

6. **Security:** In terms of security, MySQL provides a strong encryption mechanism for tablespace data. It provides strong security parameters in terms of choosing good passwords, not granting unnecessary privileges to users, and ensures application security by preventing SQL injections and data corruption.
MariaDB has had a significant boost in terms of security features like internal security and password check, PAM and LDAP authentication, Kerberos, user roles, and strong encryption over tablespaces, tables and logs.
7. **Extensibility:** A database supporting an extensible system can be extended by the user in many different ways like adding new data types, functions, operators, aggregate functions, index methods and procedural languages. MySQL has no support for extensibility. MariaDB is built on a modern architecture that is extensible at every layer—client, cluster, kernel and storage. This extensibility provides two major advantages. It allows for continual community innovation via plugins, which means that a variety of storage engines, like MariaDB ColumnStore or Facebook's MyRocks, can be integrated through MariaDB's extensible architecture. Additionally, it makes it easy for customers to configure MariaDB to support a wide variety of use cases, ranging from OLTP to OLAP.
8. **JSON support:** MySQL supports a native JSON data type that enables efficient access to data in JSON (JavaScript Object Notation) documents. The JSON data type provides the following advantages over storing JSON-format strings in a string column:
 - Automatic validation of JSON documents stored in JSON columns. Invalid documents produce an error.
 - *Optimised storage format:* JSON documents stored in JSON columns are converted to an internal format that permits quick read access to document elements. When the server later has to read a JSON value stored in this binary format, the value need not be parsed from a text representation. The binary format is structured to enable the server to look up sub-objects or nested values directly by key or array index without reading all values before or after them in the document.

On the other hand, MariaDB Server 10.2 introduces a comprehensive set of 24 functions for reading and writing of JSON documents. In addition, the JSON_VALID function can be used with a check constraint for validation, while functions like JSON_VALUE can be used with dynamic columns to index specific fields.
9. **Licensing:** MySQL offers its code as open source

under the GPL, and provides the option of non-GPL commercial distribution in the form of MySQL Enterprise.

MariaDB can only use the GPL, because its work is derived from the MySQL source code under the terms of that licence.

10. **Performance:** MariaDB achieves best-in-class performance with numerous innovations that are absent for MySQL. These include thread pool management to maximise processing efficiency and extensive optimisation features such as defragmentation within the InnoDB data store. So, when rows are deleted from an InnoDB table, the free space immediately becomes available to the operating system. There's no need to copy data from old tables to new ones, and there are no gaps in the tablespace. MariaDB also offers engine-independent table statistics to improve the optimiser's performance, speeding up query processing and data analysis on the size and structure of the tables. Without these enhancements, MySQL falls short in terms of performance. Thread utilisation in MySQL is sub-optimal and InnoDB tables become fragmented over time, compromising performance.

MariaDB vs MySQL

The following points highlight the pros and cons of MariaDB.

Pros

- MariaDB has been optimised for performance and is much more powerful than MySQL for large data sets. Elegant migration from other database systems to MariaDB is yet another benefit.
- Switching from MySQL to MariaDB is relatively easy and is a piece of cake for systems administrators.
- MariaDB provides better monitoring through the introduction of micro-second precision and extended user statistics.
- MariaDB enhances the KILL command to allow you to kill all queries from a user (KILL USER username) or to kill a query ID (KILL QUERY id query_id). MariaDB has also switched to Perl-compatible regular expressions (PCRE), which offer more powerful and precise queries than standard MySQL regex support.
- MariaDB has applied a number of query optimisations for queries connected with disk access, join operations, sub-queries, derived tables and views, execution control, and even explain statements.
- MariaDB is purely open source, instead of the dual-licensing model that MySQL uses. Some plugins that are available only for MySQL Enterprise customers have equivalent open source implementations in MariaDB.
- MariaDB supports significantly more engines (SphinxSE, Aria, FederatedX, TokuDB, Spider, ScaleDB, etc) compared to MySQL.